Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)
[Course Code: 23ECSPE1AD]

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INTRODUCTION

Cadence is an Electronic Design Automation (EDA) tool used for designing, simulating, verifying and implementing integrated circuits (ICs). It plays an important role in VLSI design by providing an environment for schematic creation, circuit simulation and physical layout design.

Cadence Virtuoso is widely used for analog and digital circuit design. It allows designers to create circuit schematics, perform simulations, analyse circuit behaviour and generate layouts. It also provides verification tools such as Design Rule Check (DRC) and Layout Versus Schematic (LVS) to ensure accurate IC design.

Cadence tools help improve design accuracy, reduce development time and support the development of efficient integrated circuits.

Applications

- VLSI circuit design.
- Schematic design and simulation.
- CMOS circuit implementation.
- IC layout design.
- Circuit verification and testing.

Tools Used

- Cadence Virtuoso
- Spectre Simulator
- Analog Design Environment (ADE)
- DRC and LVS

Comparator

WORKING PRINCIPLE

A 4-bit comparator is a combinational circuit used to compare two 4-bit binary numbers A and B. It checks whether A is greater than, less than or equal to B and produces the corresponding output. If A is greater than B, A_gt_B becomes HIGH; if A is less than B, A_lt_B becomes HIGH; and if both inputs are equal, A_et_B becomes HIGH. Only one output is HIGH at a time. It is commonly used in digital systems, processors and arithmetic logic units.

CODE:

```
module cmp_4bit (
    input wire [3:0] A,
    input wire [3:0] B,
    output reg A_gt_B,
    output reg A_lt_B,
    output reg A_et_B
);

always @(*) begin
    A_gt_B = 1'b0;
    A_lt_B = 1'b0;
    A_et_B = 1'b0;

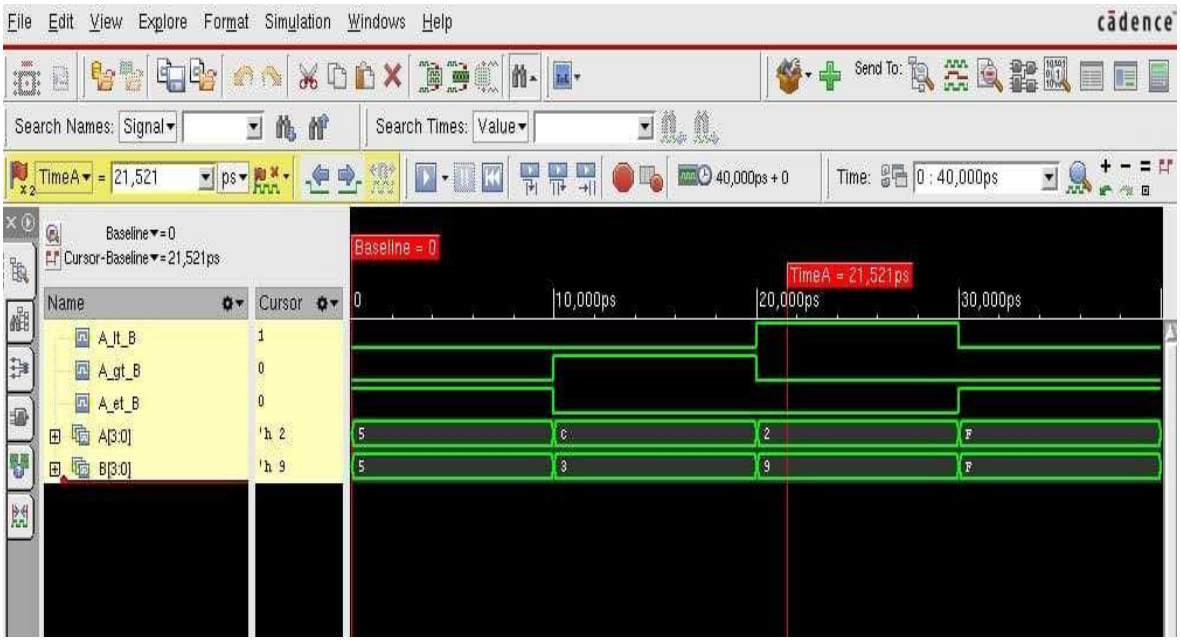
    if (A > B)
        A_gt_B = 1'b1;
    else if (A < B)
        A_lt_B = 1'b1;
    else
        A_et_B = 1'b1;
end

endmodule
```

TESTBENCH:

```
1  `timescale 1ns/1ps
2
3  module tb_4bit_cmp;
4
5  reg [3:0] A;
6  reg [3:0] B;
7
8  wire A_gt_B;
9  wire A_lt_B;
10 wire A_et_B;
11
12 cmp_4bit uut (
13     .A(A),
14     .B(B),
15     .A_gt_B(A_gt_B),
16     .A_lt_B(A_lt_B),
17     .A_et_B(A_et_B)
18 );
19
20 initial begin
21     A = 4'd5;
22     B = 4'd5;
23     #10;
24
25     A = 4'd12;
26     B = 4'd3;
27     #10;
28
29     A = 4'd2;
30     B = 4'd9;
31     #10;
32
33     A = 4'd15;
34     B = 4'd15;
35     #10;
36
37     $finish;
38 end
39
40 endmodule
```

OUTPUT WAVEFORM:


File Edit View Explore Format Simulation Windows Help
cadence
Send To:
Search Names: Signal
Search Times: Value
TimeA = 21,521 ps
40,000ps + 0
Time: 0 : 40,000ps
Baseline = 0
Cursor-Baseline = 21,521ps
Name
Cursor
A_lt_B
A_gt_B
A_et_B
A[3:0]
B[3:0]
'h 2
'h 9
Baseline = 0
TimeA = 21,521ps
0
10,000ps
20,000ps
30,000ps
5
C
2
F
5
3
9
F

Shift Register

Working Principle

The universal shift register operates based on the positive edge of the clock signal. Whenever a rising edge of the clock occurs, the circuit checks the reset signal and the two-bit selection input to determine the operation to be performed. If the reset signal is active, all four output bits are cleared to zero immediately, without waiting for a clock edge. Otherwise, the circuit performs the operation specified by the selection inputs.

The two-bit selection input, sel, determines the mode of operation. When sel = 00, the register holds its previous value without making any changes. When sel = 01, the register performs a right-shift operation, in which the existing bits move one position to the right and the new most significant bit is supplied by serial_in_right. When sel = 10, the register performs a left-shift operation, in which the existing bits move one position to the left and the new least significant bit is supplied by serial_in_left. When sel = 11, the register performs parallel loading, in which all four bits from data_in are loaded into the register simultaneously

CODE:

```
`timescale 1ns/1ps

module usr_4bit (
    input wire clk,
    input wire rst,
    input wire [1:0] s,
    input wire [3:0] d,
    input wire msb_in,
    input wire lsb_in,
    output reg [3:0] q
);

always @(posedge clk or posedge rst) begin
    if (rst)
        q <= 4'b0000;
    else begin
        case (s)
            2'b00: q <= q;
            2'b01: q <= {msb_in, q[3:1]};
            2'b10: q <= {q[2:0], lsb_in};
            2'b11: q <= d;
            default: q <= q;
        endcase
    end
end

endmodule
```

TESTBENCH:

```
`timescale 1ns/1ps

module tb_usr_4bit;

reg clk, rst;
reg [1:0] s;
reg [3:0] d;
reg msb_in, lsb_in;
wire [3:0] q;

usr_4bit uut (
    .clk(clk),
    .rst(rst),
    .s(s),
    .d(d),
    .msb_in(msb_in),
    .lsb_in(lsb_in),
    .q(q)
);

always #5 clk = ~clk;

initial begin
    clk = 0;
    rst = 1;
    s = 2'b00;
    d = 4'b0000;
    msb_in = 0;
    lsb_in = 0;

    #10 rst = 0;

    s = 2'b11;
    d = 4'b1011;
    #10;

    s = 2'b00;
    #10;

    s = 2'b01;
    msb_in = 1;
    #10;

    msb_in = 0;
    #10;

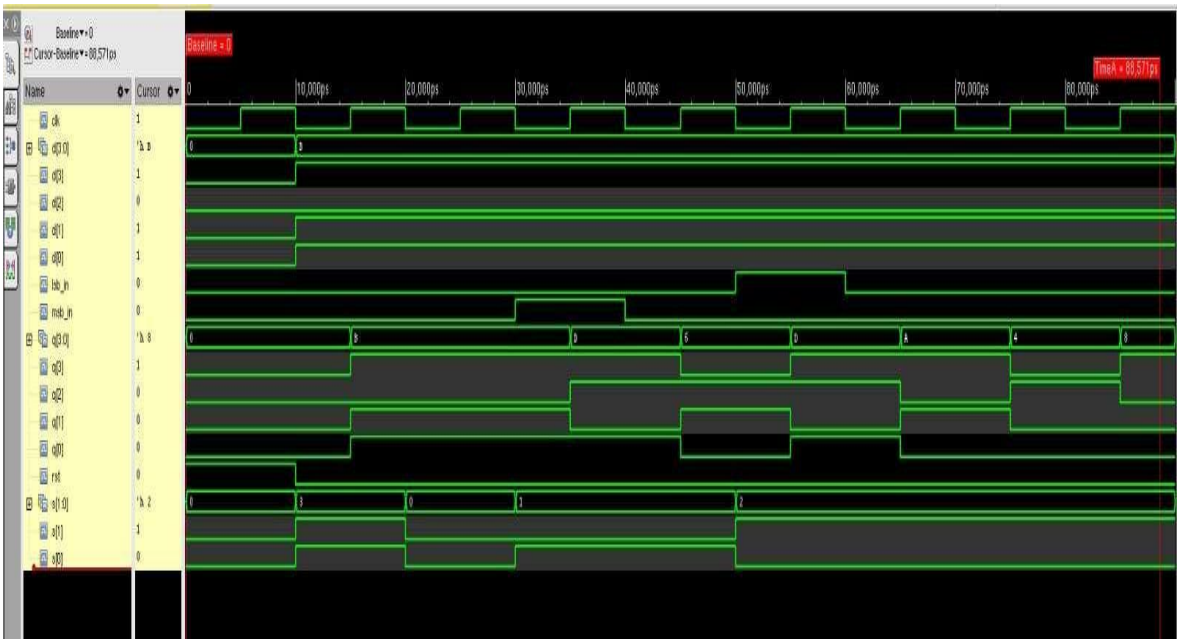
    s = 2'b10;
    lsb_in = 1;
    #10;

    lsb_in = 0;
    #10;

    #20 $finish;
end

endmodule
```

OUTPUT WAVEFORM:


Baseline = 0
Cursor-Baseline = 89,571ps
Name
Cursor
10,000ps
20,000ps
30,000ps
40,000ps
50,000ps
60,000ps
70,000ps
80,000ps
clk
d[3:0]
d[3]
d[2]
d[1]
d[0]
lsb_in
msb_in
q[3:0]
q[3]
q[2]
q[1]
q[0]
rst
s[1:0]
s[1]
s[0]

CONCLUSION

Thus, the 4-bit Comparator and 4-bit Universal Shift Register were successfully designed and simulated using Verilog HDL in Cadence. The comparator compares two 4-bit binary inputs and indicates whether A is greater than, less than or equal to B, while the universal shift register performs hold, shift right, shift left and parallel load operations based on the selection inputs. The functionality of both circuits was verified successfully through simulation for different input combinations.